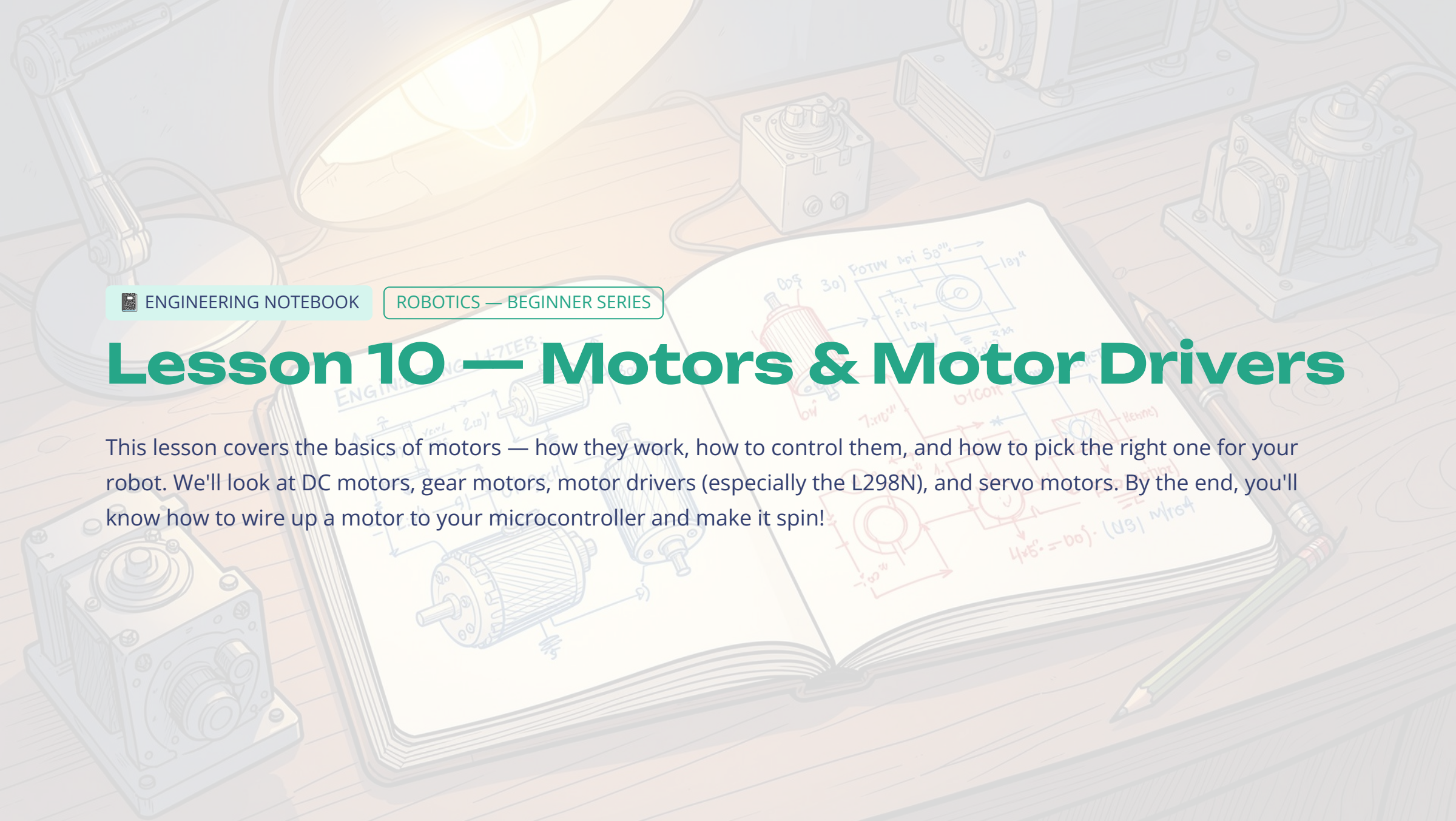


An illustration of a wooden desk with various engineering components. In the top left, a desk lamp with a yellow glow. In the center, an open notebook with technical drawings of motors and circuit diagrams. To the right, a green pencil. Scattered around are several small DC motors and motor drivers. One motor is in the bottom left, another in the top right, and a third in the middle right. A small electronic component, possibly a motor driver, is in the top center. The background is a light blue gradient.

 ENGINEERING NOTEBOOK

ROBOTICS — BEGINNER SERIES

Lesson 10 — Motors & Motor Drivers

This lesson covers the basics of motors — how they work, how to control them, and how to pick the right one for your robot. We'll look at DC motors, gear motors, motor drivers (especially the L298N), and servo motors. By the end, you'll know how to wire up a motor to your microcontroller and make it spin!

What is a Motor?

A **motor** is a device that converts **electrical energy** → **mechanical motion** (rotation). Almost every robot needs motors — they're what make wheels spin, arms move, and grippers open and close.

Why Robots Need Motors

- Locomotion — wheels, tracks, legs
- Manipulation — arms, grippers
- Actuation — valves, pumps, fans
- Precision — positioning, steering

Energy Conversion

Input: Electrical energy (voltage + current)

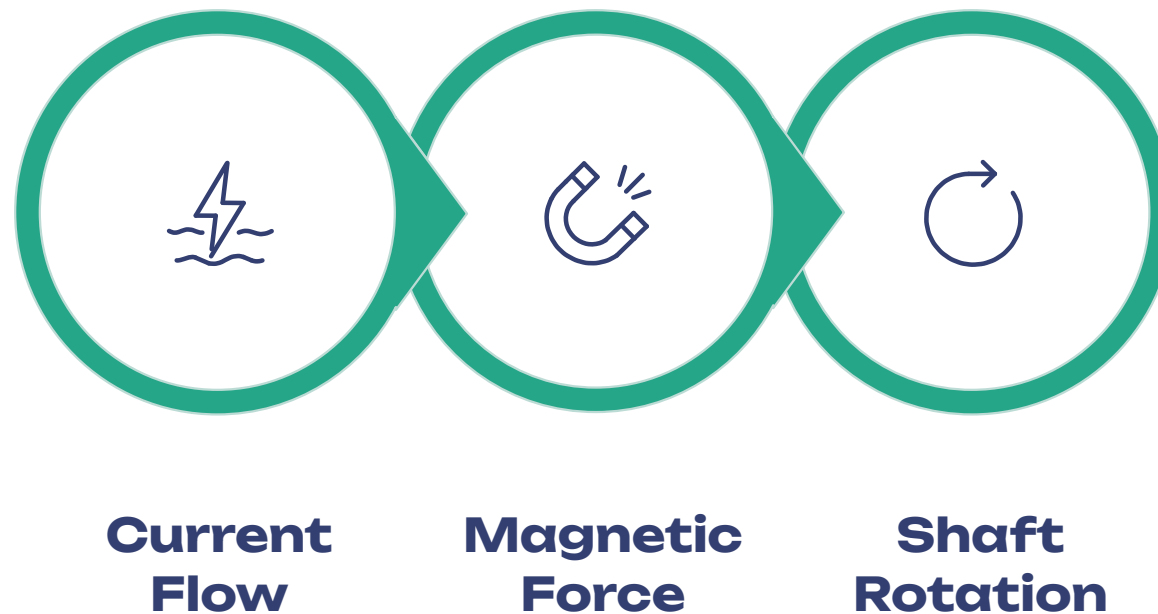
Output: Mechanical energy (torque + RPM)



★ Key idea: More current → more torque. More voltage → more speed (roughly).

DC Motor — Working Principle

A **DC motor** works on the principle that a **current-carrying conductor in a magnetic field experiences a force** (Lorentz Force). This force creates torque and makes the shaft rotate.



The diagram above shows the basic sequence — current through the coil creates a magnetic interaction that produces rotational force on the shaft.

Direction of Rotation

- Determined by **polarity** of voltage applied
- Swap +ve and -ve → motor reverses
- Use an **H-Bridge** circuit to reverse electronically

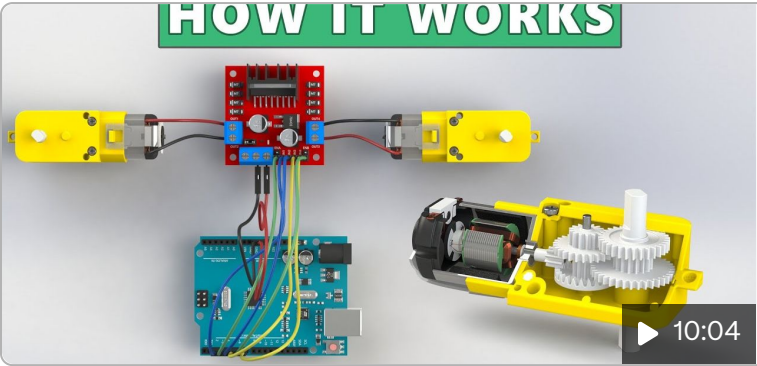
Speed Control

- $\text{Speed} \propto \text{Applied Voltage}$
- Use **PWM (Pulse Width Modulation)** from microcontroller
- Higher duty cycle = faster rotation



Simple sketch idea: Draw a circle (motor body), two terminals (+ and -), and an arrow showing shaft rotation.
Label: Armature, Commutator, Brushes, Shaft.

DC Gear Motor



 YouTube

How DC GEAR MOTOR Works with ARDUINO and L298N

In this video you're going to learn how DC gear motor works, and how to use gear motor speed control with L298N. This is a beginning of a fun experience on how...

A **DC gear motor** is a regular DC motor with a **gearbox** attached to its output shaft. The gearbox trades speed for torque — super useful in robotics!

What Are Gears?

Interlocking toothed wheels that transfer rotational motion.
Different sizes = different speed/torque.

Speed vs Torque

Gear reduction **lowers RPM** but **increases torque**. Perfect for moving heavy loads slowly.

Robotics Uses

Robot wheels, robotic arms, conveyor belts, elevators — anywhere you need **force**, not just speed.

Gear Ratio Formula

Gear Ratio = N_2 / N_1

Where N_1 = input gear teeth, N_2 = output gear teeth

Example: 40-tooth output ÷ 10-tooth input = **4:1 ratio**

→ Output speed = 1/4 of input speed

→ Output torque = 4× input torque

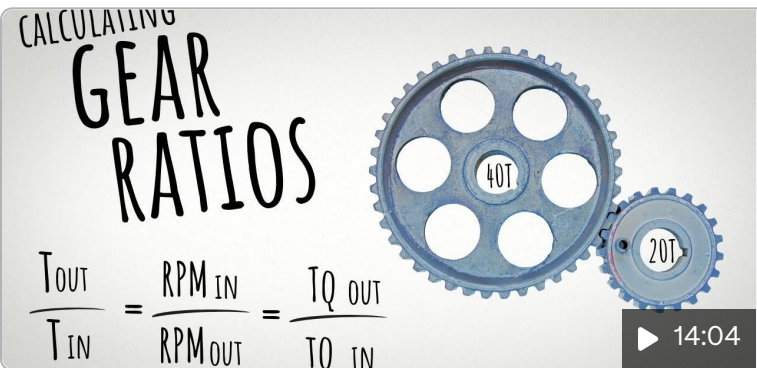
★ Quick Example

Motor: 1000 RPM, 0.5 N·m torque

Gear ratio: 5:1

Output:

- Speed = $1000 \div 5 = \mathbf{200 \text{ RPM}}$
- Torque = $0.5 \times 5 = \mathbf{2.5 \text{ N}\cdot\text{m}}$



 YouTube

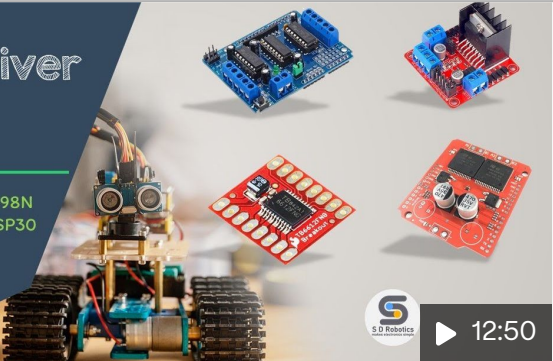
How To Calculate Gear Ratios: Torque, RPM, & Mechanical Advantage

Learn the fundamental principles of gear ratios to calculate torque, speed, and mechanical advantage for any project.  Why Watch? Ever wondered how a...

Motor Driver — Why Do We Need It?

Motor Driver In Depth

H-Bridge • L293D • L298N
• TB6612FNG • VN12SP30



YouTube

Motor Driver in depth | H-Bridge | L293D, L298N, TB6612FNG, VN12S...

Motor drivers are a very important component when working with robotics or mechatronics project. So in this video, we will discuss the details about motor...

12:50

You **cannot** connect a motor directly to an Arduino or ESP32. Here's why — and what to do about it.

⚡ Current Problem

Arduino GPIO pins can supply only **~40 mA max**. A small DC motor needs **200 mA to several Amps**. Direct connection = fried microcontroller! 🔥

↺ Direction Problem

GPIO pins can only output HIGH or LOW — they can't **reverse polarity** to change motor direction. You need an **H-Bridge** circuit.

🛡️ Protection

Motors generate **back-EMF** (voltage spikes) when they stop or change direction. Motor drivers have built-in protection diodes.

⚠️ **Never** connect a motor directly to a microcontroller GPIO pin. Always use a motor driver module!

L298N Motor Driver — Pinout & Connections



YouTube



Speed Control Tutorial of Single DC Geared Motor With L298N Motor ...

Here's the video, showing the process to control the speed of a Single DC Geared Motor with Arduino UNO(same connection & Coding for Arduino NANO also) &...

The **L298N** is a dual H-bridge motor driver module. It can control **two DC motors** (or one stepper motor) with direction and speed control. Very popular in beginner robotics!



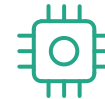
Motor Supply (Vs)

5V – 35V DC. Connect your motor battery here. **Not** the same as logic power!



Motor Outputs (OUT1–OUT4)

OUT1 & OUT2 → Motor A. OUT3 & OUT4 → Motor B.
Polarity here determines rotation direction.



Logic Inputs (IN1–IN4)

Connect to Arduino GPIO. IN1/IN2 control Motor A. IN3/IN4 control Motor B. HIGH/LOW combinations set direction.



Enable Pins & GND

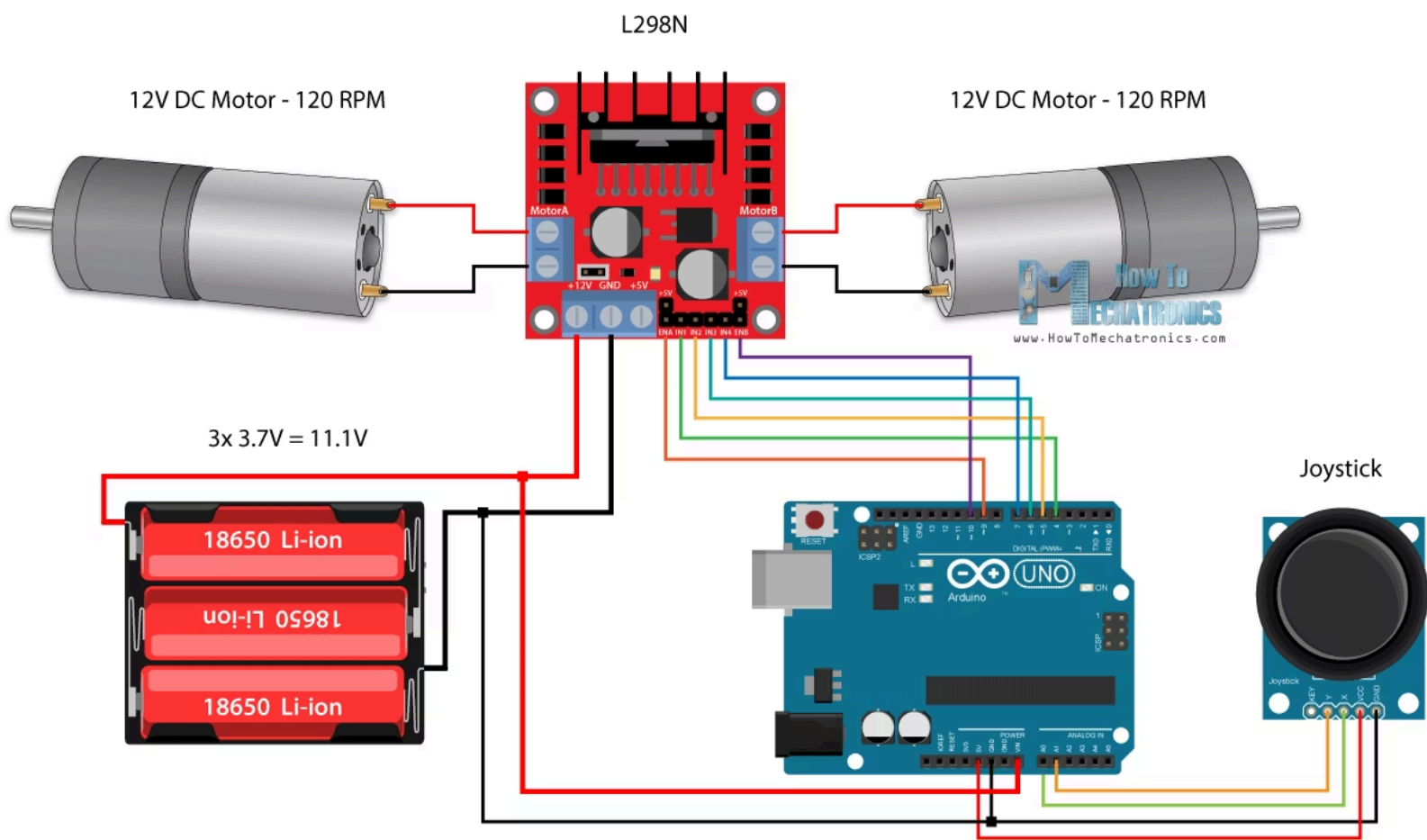
ENA & ENB: Connect to PWM pins for speed control.

Common GND between Arduino, driver, and motor supply is critical!



Wiring Diagram — Arduino + L298N + DC Motor

Here's the complete wiring setup. Follow this carefully — **common ground is a must!**



Connection Table

Arduino Pin	L298N Pin	Purpose
D9 (PWM)	ENA	Speed control
D8	IN1	Direction A
D7	IN2	Direction A
GND	GND	Common ground ★
5V	+5V (if jumper on)	Logic power

Power Wiring

- **Motor Battery +** → L298N Vs
- **Motor Battery -** → L298N GND
- **Motor Wire 1** → OUT1
- **Motor Wire 2** → OUT2

★ **Common GND rule:** Arduino GND, L298N GND, and motor battery GND must ALL be connected together!

Direction Control & H-Bridge Concept

The L298N contains an **H-Bridge** — a circuit of 4 switches (transistors) that can reverse current direction through the motor electronically.

IN1 / IN2 Truth Table (Motor A)

IN1	IN2	Motor State
LOW	LOW	 Stop (brake)
HIGH	LOW	 Forward
LOW	HIGH	 Reverse
HIGH	HIGH	 Stop (brake)

H-Bridge Visual

Think of 4 switches arranged like the letter **H**:

- Top-left + Bottom-right ON → Forward
- Top-right + Bottom-left ON → Reverse
- All OFF → Motor coasts to stop

The L298N handles all this internally — you just set IN1/IN2!



Speed Control with PWM: Keep IN1/IN2 fixed for direction. Vary **ENA** using `analogWrite()` (0–255). Higher value = faster motor.

Servo Motor — Position Control



YouTube

Servo Motors, how do they work?

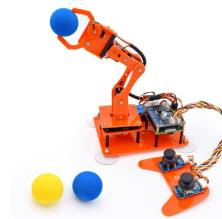
Servo motors explained, how do they work? Exclusive VPN offer from \$1.98/m 👉
<https://www.privateinternetaccess.com/TheEngineeringMindset> Download...

A **servo motor** is different from a DC motor — it doesn't spin continuously. Instead, it moves to a **specific angle** and holds that position. Great for robotic arms, steering, and precise movements!



Basic Components

- DC motor (inside)
- Gearbox (for torque)
- Potentiometer (position feedback)
- Control circuit (compares target vs actual)



Robotics Applications

- 🤖 Robotic arm joints
- 🚗 RC car steering
- 📷 Camera pan/tilt
- 🚪 Flap and valve control

How Position Control Works

Servo receives a **PWM signal** on its signal wire:

- 1 ms pulse → 0° position
- 1.5 ms pulse → 90° (center)
- 2 ms pulse → 180° position

Arduino **Servo** library handles this automatically with `servo.write(angle)`.

Servo Wire Colors

- ● Red → VCC (5V)
- ● Brown/Black → GND
- ● Orange/Yellow → Signal (PWM pin)



Motor Selection Checklist

Before buying a motor for your robot, run through this checklist. Picking the wrong motor is one of the most common beginner mistakes!

1

Required Speed (RPM)

How fast does the wheel/arm need to move? Calculate from desired robot speed and wheel diameter.

2

Required Torque (N·m)

How much force is needed? Consider robot weight, friction, and incline. Add a 2× safety margin!

3

Supply Voltage

Match motor voltage to your battery. Common: 3V, 6V, 12V. Don't exceed motor rating.

4

Current Draw

Check stall current. Your driver and battery must handle it. Add a fuse for safety.

5

Load & Weight

Heavier load = more torque needed. Gear motors help, but don't overload the motor.

6

Gear Ratio

Need more torque? Use a higher gear ratio. Need speed? Lower ratio. Balance both!



★ **Lesson 10 Summary:** Motors convert electricity to motion. Use gear motors for torque. Always use a motor driver (L298N). Control direction with IN pins, speed with PWM. Use servos for precise angles. Check the checklist before buying!